

ActInf GuestStream 080.1 ~ Laura Desirée Di Paolo: "Active Inference Goes to School"

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all right hello it is April 22nd 2024 we're in active inference guest stream 80.1 with Laura desire Deo talking about active inference Goes to School the importance of active learning in the age of llms we'll have a presentation than some discussion so thank you Laura for joining looking forward to this thank you very much for having invited me um okay so I'm going to present practically one of our latest publication which is um in fact that the inference Goes to School uh oh wait I have to Yes um this is an overview of the talk more or less I'll try to go as fast as I can for the first part because I'd like to spend a little bit more time on the later slides and I'm sorry for some misspelling or mistakes that you will find here and there in the in the images because I had long fights with Jud for trying to uh um generate images that would vate better my my my talk and at some point I after 50 or 60 uh attempts I simply gave gave up and they choose just the best one that could I could uh I could achieve to I could achieve okay so ah still I have to use my mouse uh just few things about myself I've been uh interested in learning since I uh uh I started University maybe because I was uh considered a slow learner in uh in high school definitely not an high achiever uh um and therefore when I started to uh to study at University I started to study learning different kind of learning different learning across species and the evolution of learning and so on and so forth uh till the point in which I started to be interested in learning in educational settings thinking okay this is the perfect recipe for Success because you have all the ingredients that you might have you have super curious beings for whosoever had to deal with children before they uh go to school uh knows that they um have this insane curiosity and they ask wise about everything theoretically in school you should have tons of learning opportunities and you finally have people that are trained to give uh the answers that the children seek to have unfortunately the sit the situation in school is not always so nice for quoting the um principal a British principal and during an interview who said it's incredible that children prefer to be home sick that going to school the place in which they can learn stuff consider that learning is fun and for what concerns me I am

a researcher and as I assume most of us um I love learning I think learning very very fun so I feel part of my my job or my duty to bring back the fun into learning particularly in educational settings or in compulsory education um this is just a slide with some uh basic literature about active inference and learning I won't be too specific about active inference uh there is here a that eventually could um could follow up for some questions more specific about the active inference because it's way more prepared than I am but anyway this is like um I don't know if you have I have to move this one okay perfect um uh this is a basic model of active uh inference the idea of cognition in active inference for which the agent and the cognitive system of the agent is not waiting for the information to arrive to fill in but instead uh act proactively so makes predictions about the state of the world in which expect to leave the information that that expect to uh to perceive into the world and then every time compare the predictions that has made with the effective information the effective situation of the world in that particular moment and when there is a discrepancy discrepancy uh in in terms of both surprise or events that are unexpected or errors uh either update the model for making better predictions or change the words and so act onto the world for uh the original predictions in this model of cognition learning is a sort of imperative because every time that there is a discrepancy between the prediction and the state of the world is like a red flag that says okay here there is something that can be learned there is something new that can um can be acquired for making my future prediction my future and error minimization better um another interesting aspect of active inference is that predictions are not just in terms of thoughts beliefs or knowledge for instance but also in terms of sense of motor activities so everything that we perceive feel see um is considered a prediction about the world and it's considered in itself cognitive a way to understand the world that surrounds us the model of active inference um of learning and cognition together is very efficient and is uh very powerful for explaining for instance different kinds of uh learning or learning mechanism using the same very simple so to say uh mechanism you can um explain for instance um individual trial and error and imitation as every time the generative model that optimize on the environment and on the situation the environment for acquiring the the right information or for making the the the best predictions possible um you can also explain how expertise in terms of refined knowledge what is called know what uh or in terms of skills uh can very easily come from explorative policies and for Curious Behavior into the world um also importantly focuses on the role of attention that if you think in terms of educational settings and educational environment is of course very important again in terms of prediction error uh Dynamics and surprise um for the acquisition

of competencies and skills and most importantly at least for me using the inference to uh process that very often are considered complementary but separated which are development and learning they can be Rec combined together because making predictions particularly in terms of sense of motor activity and sense of motor interaction with the word means also acting on our biology and neurobiological structure for being sure that these predictions are as accurate as they can be okay when we wrote said this when we wrote the paper we had three main problems the first problem was how do we apply a theory of cognition such as the active inference in real educational practices and conversely how these educational practices should look like for fitting our theory of cognition on these two problem adds up a third problem which is the uh the use and the spread of artificial intelligences particular L large language models in education as well as in other uh in other fields and how this might change the rules the problem of application of um large language models particular artificial intelligence or even other form of Technologies within educational settings but particularly uh large language models um is given uh mainly by the fact that they are disembodied when a user has to deal with a um larger language models in reality deals with some artifacts the computer the the screen the the keyboard and so on that can touch see and perceive but at the end of the day um um Deals Only with the input output series and the whole process all the other artifacts that are imprinted that are inside the uh the model itself are completely uh invisible um which makes a problem because normally we are used for instance we can't make Behavior reading or understanding intentionality behind the answers of uh of chpt now I added uh yesterday morning this uh this slide this is a beautiful paper that came out I think two or three days ago about uh children the P the perception of children about information on Google and uh it's very interesting because it somehow it it has the same problem that um that llms have uh children even very young children trust the information that Google uh gives but they don't have any um any possibility to understand where this C trust come from in the moment in which they have to deal with something like a chpt in which the the answer is given in a sort of natural language of course the problem is even uh is even deeper uh therefore how do we nurture the human generative model in in the a in the in the age of the artificial intelligences um these pictures that I have here on my slide I was looking around for uh different representations of how integrate artificial intelligences within educational settings uh and they all are accumulate um they have one thing in common which is they tend to treat the artificial intelligence in in a sort of human way because this is if the artificial intelligences are disembodied um it stats that we are embodied embedded extended and um enacted agent so

we are uh um um active inference agent so our generative model for understanding for dealing with these embodied entity entities as somehow to um to put them in a way that we can understand so our problem would be how do we build an educational environment in which our generative models can effectively and uh um efficaciously deal with the powerfulness of these uh new technologies one way is what I call here the traditional way this is the only image that I got from jpt that was perfect at the first attempt I asked jpt to give me the image of a classroom with poop pills between eight and 10 years of age um the fact that jpt gave me directly this image tells me a lot because jpt is trained on our human knowledge and this is very likely the image that we all have when we think about a classroom about education this that I called here traditional uh traditional educational setting in reality as a specific name which is the teacher centered approach which I have somehow schematize here um and more or less works this way you have all the human knowledge that is far back in the background of the educational setting itself this extensive human knowledge is paralyzed and organized in curriculum that then affects the child on the other hand you have a teacher that also affects the child and you have this child that somehow acquires these information and grows up having been filled by the information that is provided with um somehow scarily this kind of uh structure is very similar to the kind of training that we do for disembodied entities such as llms with the additional uh problem that the fine tuning in educational setting in these kind of structures uh happens at different level not only the child is fine tuned by the teacher and by the curriculum having somehow knowledge that exists other knowledge outside of the classroom outside the knowledge that is provided within the educational settings uh but the fine-tuning works on the teachers themselves and the teachers themselves they don't fine tune every single child but they usually fine tune our old classroom now for whosoever knows how the fine tuning for instance in L works um knows that the more you fine tune a model the least creative the model is going to be now um when we think about this kind of traditional settings uh and we think that the active inference is the best um uh model of cognition that we currently have uh we can't uh avoid to uh to uh recognize that it works also in these settings problem is yes it works but is not really on point in the sense that um for instance the uh um very often in this kind of settings we pretend that the attention of the poops goes specifically to certain uh uh certain um feature of the environment the the the lessons that are provided by the teachers uh while in fact we know also that children at particularly in early age so till round about 12 they tend to uh pay attention way more and to learn a lot from irrelevant contextual information which in active inference terms makes a lot of sense because is a

generative model that that at uh till that point in time very few experiences so is gathering information in the environment for making a better prediction on the long term another problem with this uh uh with this model and this is the only concept that is not in the paper that I have added in the slide is what I called here error learning Dynamics um another problem with the this kind of settings is what we consider the error signal we have said before I have said before uh in the moment in which there is a discrepancy between the prediction that the cognitive system has made and the effect the actual uh um environmental uh situation there is an error signal if the error signal doesn't come exactly from the mistake but comes from for instance from the feedback that might arrive in terms of notes or from being scolded at home because you get a bad grade or you you got a A not in school then of course the change of the state the change of the situation the acting onto the world and the update of the generative model and of the prediction H doesn't fit with the uh um uh with the uh with what we want the children to learn uh these studies that I have here uh quoted some of them are really beautiful studies of Solange dero uh she has observed different children in different educational settings and also children and adults how do they they react to mistakes and how they monitor mistakes um and discovered that children educated in this kind of traditional or teacher centered uh approach uh educational settings develop more than uh the ability to learn from mistakes develop what is called error avoidance which again makes sense it's not that the the active inference model doesn't work in this way it's only that the error signal doesn't come from the mistakes for instance in the task or in the exercise that the child has made but comes from another aspects of their world okay therefore how should the classrooms in which we want effectively to Foster an idea of an active inference agent to nurture an active inference agent um it has to be an educational setting it has to be a classroom that offer um um explorative opportunities and offer the time for precision waiting uh what I mean with that um it has to be a educational environment in which the um there is the possibility to learn from uh various aspects of the world and there is there has to be the possibility for the for the child for the child to um fine-tune its generative model uh multiple times on the same task in a way that can really really really uh understand um at best how to perform it it has to be an environment in which there are u in which it is fostered also a multimodal and multisensory way of approaching learning and knowledge because as we said the um predictions are not just in terms of know what the the um um information that is provided to the child but also all the irrelevant stimuli that are around so irrelevant so they shouldn't be irrelevant should be instead relevant and has to Foster also individual interests and attention modalities because at

the end of the day the generative model is always individual it's always the the individual that uh update its own generative model in its own specific environmental settings Al there are of course some coach forces as AEL would call them that makes us being able to um act and to think in in terms of our community of our culture it has to be an educational environment in which development and learning are considered together and not separated in these traditional educational settings for for instance we have all the classroom that very often are divided by age not about skills or stage of development um it has to be an environment in which uh the construction of uh obstruction and categorization has to come through environmental experiences which again in activ inference terms makes a lot of sense constructing abstract uh knowledge constructing abstract categories uh allows the system to be somehow prepared to how to deal with certain subject or with certain object or with certain circumstances uh and then requiring just the fine tuning of the of the generative model on the specificity of a particular circumstance and it has to uh to uh be an environment in which the ER learning Dynamics are considered in a way that error signal comes directly from the error and not from from other um contextual uh information um instead of thinking to invent from the from the top of our mind a uh um a new educational approach we looked around to um search for which educational approach that's already out there that offered the offered the most opportunity to fit with our idea of educational settings uh among various the monor meod is the one that if doesn't f fit all the boxes uh definitely fits most most of them um for describing uh a little the in the monor meod you will always find a sort of open spaces and multi-age uh environment of learning curricula are always spanned on free to six years um and not on just one year as in um in traditional schools uh children are always free to choose what to study for how long uh with whom uh it it is already implemented within the PED the pedagogy and within the uh the the theory also of mar monor the idea that Learning and Development are just one process it works through what is called an indirect form of teaching and through the prepared environment prepared environment is the classroom as is called in the in the monor method um which is already organized in a way that can uh Foster some kind of not not not only of skills but also of uh construction of specific knowledge and pays a lot of attention to the error Dynamics uh uh through the design of the classroom itself but also through the design of the specific learning material I have some examples in in a couple of slides um the materials in a way that the material is is design so the material is designed in a way that are poop particularly in the lower uh stages so in the in the first grades uh can immediately perceive when uh they have made a mistake and um Can immediately correct them so the error signals uh arrives

directly it doesn't there is no delay um and the old classroom is organized in a way that the uncertainty so the the um prediction minimization is somehow manageable and still organized in Progressive diff difficulties but every time adding up just one extra variable one extra element of uncertainty now is well known I wanton if anyone wants to know more about the the history of maror uh I can talk about it later but it's well known that the method was invented by maror Italian physician at the uh in at the beginning of last century I have been uh lately dealing with the most applied of our work particularly in mathematics in geometry and in um grammar and I have to admit that she was frankly genial uh the the the design of her material is is is astonishing still today um Deon is incredibly famous and Incredibly well known there are more than 15,000 montori School in the world uh in more than 140 countries uh in 2011 um came out a paper uh in business psychology I think it was called the monor mafia because some of the richest and most well-known entrepreneur in the world um are orw montor poils um but still it's not so much well known the theories of development and learning of the monor meod are not so well known for instance in cognitive Sciences so they are very much I was very surprised the other day looking at the how many results one can have putting mono and entrepreneur uh in on Google Scholars uh and not so much in cognitive Sciences this is not really related to the topic but is also a good aspect of the montor method which is given the particular attention that gives to the ecological context for learning it is also it has also a leading position in terms of ecological sustainability within the educational setting and within the curricula and this is of course a topic of major interest that should in my opinion at least be part of whatever C in whatever School independently of anything else okay um these are some examples of montesorri schools the montori schools are more or less all structured in similar ways sorry uh in similar ways um first and foremost there is no distinction between inside and outside classroom uh every space given the physical constraints of the buildings uh might be corridors might be um small or big rooms the doors are always open the children and the poils are always free to move around there has to be always a free a big free space available for uh children that prefer to work on the floor and also because some of the materials requires lots of space there is the material for history the history band that's up to 5 m long and is designed exactly for the children to understand to have a a um a perceptual understanding of the different um positioning on the on in time of the different characters or events that they are working on the inside and outside as I said they are both um educational into sense not only because there are lots of activities like the observational nature and part of the curriculum of uh botanics biology and so on that are literally run outside but because the

old FURNITURES and uh the all everyday items are easily portable outside and very often um you might find class the classrooms or the normal uh daily work happening in in the outside space uh there is one of the image that I have here in in black and white uh taken from from the B of from the book of Schneider and Mueller refers to a mono School uh in Germany in the 30s um so it's historically designed in this way uh within the montori classroom also the construction of abstract uh category ories and of categories in general is um is supplied by the uh by the environment itself uh through the positioning of the objects of the materials the books that refer to specific uh disciplines of specific subjects uh that are positioned in a way to construct more and more abstract categories for instance you might find one might find the materials for uh botany and materials for uh chemistry and the materials for uh physics all under the label science and each one of them will be labeled as well in a way that children simply interacting in in with the space interacting with the material and with the material and simply with the simple activity of taking the material out from the shelf and put it in back understand more or less how the different subjects are related to one another Plus for instance through the presence of kitchens that are used for making food I used for uh experiments in chemistry I used for taking care of plants and I used for uh washing the brushes allows kids to uh make um connections across different disciplines um when I talked before about error learning Dynamics and how the the error signal should be um should be uh um should immediately get back to the system for having an effect an an effective uh update of the model or modification of the word uh in terms of uh effective learning um this is one example this is actually the first material for mathematics that is in the Mont meod that's usually presented to Children um between the age of 34 uh in Italian they are called the numerical roads in English they are called the red and blue uh roads and are these uh long pieces of of wood divided in a section that um uh represent units uh divid in different colors so in alternative colors blue and red um and uh they are from 1 till till 10 with the 10 that's 1 meter long and the one that's 10 cm long um these roads um have the AIM to um help the the children to um learn the basics of uh addition additions and subtractions um making perceptual evidence evident uh how the different quantities are related to one another most importantly they don't have the AIM to uh um teach the quantities because children of three or four age they already ask for free Bone bonds instead of two so they already have an idea of quantities these kind of material is designed exactly for um uh uh making clear how quantities are related to symbols and how quantities are related to one another this is the first of a set of free materials for the uh for the the beginning of the mathematical the curriculum of mathematics which as a um as a matter of

fact has this representational a the other two instead that are the spindle B and the numerical coin uh the spindle box has instead uh like sort of the opposite aim in this case the child has in front of them a box with already uh some uh the serious of number uh numbers impressed on the box and with 45 spindles all identical 45 $9 + 8$ plus seven Etc and the the work consist in taking the spindles out from the from the box and putting them back according to the in in a way that the number of spind of of spindles in each um in each compartment has to um to be the same that is represented uh by the symbol itself um as we see here the series of number goes from 0o till 9 in this particular exercise compared to the other one we will have just one extra element which is the child has to understand that in a series of number zero represent no quantities so it has the same so it somehow reinforces the uh the knowledge that has previously acquired with the other uh with the other exercises and now adds adds up with an over element the third uh exercise in m ma matics is the numerical coin in this case that it works as sort of test for children of three four years of agage in which uh the children have not only to combine the symbols with the quantities but also to put the symbols in the right order of course there is no shame in going back and in doing the different exercise at different levels and this is this is very important because as we uh see here this um I'm sorry that I Ed the Italian but after having had the very bad experiences with the the English translations I now uh I tend to use directly the Italian um uh this this page refers actually to the different uh calculations that can be uh done using this kind of uh this kind of material at different stage of development if a child of three or four um might do the basic arithmetics of $9 + 1$ $8 + 2$ Etc noticing how um $9 +$ one so um combining the road of nine plus the road of one you have the same uh measure the same Dimension that the road of 10 so it's equal 10 a child a little bit uh older that had already some experience with other forms of calculation for for instance multiplication so my notice that uh combining the uh the roads you might uh have uh five roads all with the same the same length the length of the 10 roads and you will have left the five roads in this way making a different calculation so calculating the total um number the the total quantity total uh number of units in this series which is 10 multiplied five + five which at the end gives M the possibility at again a different stage to calculate so gives the basics for calculating the um complete sum the complete quantity of uh um the complete quantity within a series of given numbers okay and I swear I am about finish these are uh three of the three other um elements of the extensive curriculum in mathematics and science it's not a case that the motasi um method seems to have the best result in mathematics and science because the the curricul the materials for mathematics and science particularly for the um the first grades is

the most extensive one um now I'm not describing in in detail all these material just the the pearls that are the one on the on the left of the screen are used for making a little bit more complex calculation and also for teaching to uh visually teaching the uh the poils how to calculate cubes volumes um but some times ago I uh attended to this beautiful uh talk of Anthony trory that is also an implementation in school for the study of Computer Sciences in in primary education um through a concept that is called the concreteness fading so these are implementation for understanding how to uh help uh poops particularly young children to construct abstract knowledge particularly for very abstract uh for very abstract competencies such as uh computer science uh the computer The concreteness Fading approach practically consist always in three moments for which we start from a concrete element in this case is the dead man island is is a sort of game that the the kids have to perform an intermediate phase in which you add uh to uh some some elements of the concrete phase some elements that will be in the more abstract phase for phase for then finally end up with the obstruction this kind of concreteness fading approach is everywhere in a mono classro and it has the advantage that is not just an implement mentation because the implementations are useable and beautiful the problem is that once the implementation is brought away we have the classroom as it stay as as it was before instead in this case this kind of um this kind of uh this these elements are present in the classroom at all time and this also helps for um uh um somehow solving some problems with this approach to uh for helping children in constructing abstract knowledge which is as always in in science in literat particular in in literat education some people think that the best way to help kids in constructing abstract know knowledge is going from concrete to abstract some people think exactly the opposite to present first the abstract the abstract um knowledge and then to allow the kids to make connection between this abstract knowledge to the uh to the uh concrete elements in in um in everyday word um we can assume very easily using again an activ inference point of view that both approaches might be right very much depends on the kids having this physical elements all the time into the environment allow different kids with different generative models to construct their own um their own knowledge in their own way some kids might find easy to go from concrete to abstract some kids might find easy to go from abstract to concrete some kids might find easy to go from intermediate to abser concrete but anyway they will always be surrounded by the same objects for extensive periods of time and this allow also to uh refine and to maybe uh going from one to the other Approach at different stages of development um now the mon of course is not the only uh valuable method of Education that exists in the world this is one of the the ones that I I like the

most it's called the learning by the sign it has a different um different voices in in educational literature uh this is just one of them and I particularly like because uh things to um um change or to improve educational settings not from the head but from the feet uh so not through um educational pedagogies but really and physically rebuilding the uh environments in which the children or the poop learn every day the problem with this and other approaches of this kind is that they are not particularly precise in my opinion in looking at the error learning Dynamics and if we don't pay attention to uh um uh what a a poil perceives as an error what is gives the the error signal to the generative models of course the learning is way more difficult and this way more uh more complicated and at the end of the day when we want to design educational uh um educational environment we want any way to transmit certain kind of uh cultural and uh cultural competencies and the cultural knowledge finally um um we also haven't dedicated much at the end much thoughts about llms the the point with LMS is that it's clear that they are here to stay and is somehow recognized that they will have a huge impact on education the problem is how do we Implement them if we Implement them in a sort of traditional educational settings they will do for the poop what teacher teachers do which means they will provide knowledge but we know that the knowledge that is provided by llms is not always accurate is sometimes messed uptime is mixed up sometimes is is clearly invented therefore it's if we want to really uh use them within educational uh settings and we should because they are powerful uh entities um we should do that with what kasal have find a clear educational approach pedagogical approach uh with a strong focus of critical thinking and strategies s something as the montor meod finally the idea that I have that we have is more or less to combine the elements some elements that are taken from the montor from the Mont meod particularly for what concerns the uh the application in educational settings because this this method already has in Focus uh not only the child but also the teacher and also the curricula which is always um somehow regenerated by the interaction between the child and and the teacher is I thing maybe I should have mentioned earlier in in the montori in the montor meod the um interaction between teacher and poils is is seen as a form of collaboration not as a form of giving some kind of information and also that as in the focus in in the center of the focus at least a part of the extensive human knowledge that we have created in Millennia compared with um uh combined with an approach of learning to learning and uh to cognition which is given by thetive inference which is a a predictive approach to uh um the relationship between a cognitive system and its environment which in my opinion at least might be really the game Cher in education when we will stop to think about school as a as a place in which the children

are and the poop are filled up with information but a place in which they will um they will try out what they already know and they will refine their knowledge uh this will be uh really a I think a game changer in in educational settings and that's all let me just thank uh Ain carlut Ben White Axel constant Andy Clark of course are part of my team at the University of s Sussex the chat lab the children and Technology lab SE at the University of s Sussex with the wonderful Nicola U and the Escape material Minds year C grant that pays my salary and allows me to uh uh work on different aspects of materiality for which I can work uh from educational aesthtics to um to Tool making activities in early home this is uh this is really fun thank you very much thank you Laura awesome talk cool all right I can stop sharing the screen on here all right thank you um to kick things off AAL if you want to give any first Reflections and questions and then meanwhile if people have any questions in live chat I'll look for that yes thank you Danielle so thank you talk I it's clear it works good thing to have in your talk uh something I wanted to say in addition resonance with what you said is that you were my opinion very correct point out that uh like an easy way to do things would be to say that activ fance tells us that we need to learn in know certain active way that is not what is happening AC inference is AAL model of cognition if it's true cognition is predictive that's it and whatever we can do will not change that um so AC inference does not tell you that mon is better or that theal system is unnatural that is not something that happens but um what you did and what I'd like to dig uh well in the general same intuition but you know different framework way of thinking is that while it does not tell you that one way to learn is better it does tell you that well learning is contextual like you if you learn to predict things you learn to predict specific things from specific stimuli so the way that you will uh learn things will educate what type of stimuli you give attention to and what type of meaning uh you give to and uh the like the last 20 400 years uh they were characterized by a very very strong cultural selection pressure for institutions that could control people doing things at very large scale so mostly nation states mostly with war and so they have a very strong interest to for people to understand things through the basically what the high things of it and it happens that ucation system Trends you to worry not about what it is that you're doing but what the higher up thingss you should think about what's doing and if we are if we are lucky uh possibly we may uh have what is already occurring I guess um well that is more and more centered around like problem solving we we we are a city we need to do something with seer what should we do and this is an open question and your hup is not like uh simultaneously a sociologist uh Way Treatment manager physicist so you'd have to have some kind of active Demar of trying to

find out what it is that we should do and the mon does that it does uh teach people to like okay you have a problem here is the environment new problem is set go ahead do what you want and this trains people that are I guess I hope more efficient in problem solving that people who are basically trying to learn the formula and repeat it and then perform the repeating formula so I think there is a pretty deep thing to be the that and I think you've identified it and I think that there is like it's interesting to think of it in terms of active inference and there is more way to go uh in understanding what precisely in terms of active Inference um sorry in by there is more work to do in understanding in terms of inference how precisely montori or classical school system shapes uh the way that people think and organize can I can I answer yeah uh I completely agree with you obviously but um and maybe I wasn't PR enough so the point is that as you said activ if this is the way in which if this is the way in which our cognitive system works it will work in the Montes meod in The Valor Shou in in whatever system it will be the problem is how much in the moment in which our intent is educative how much we are precise in thinking the environment and the precise context of learning for being sure that the poops will acquire the specific information that we want them to acquire which doesn't mean uh that we want all the kids to think the same but we want for instance to transmit a certain level of basic competencies basic mathematics basic literacy right from which then they can build up by themselves for doing that I called them here because I didn't have a a better term error learning Dynamics because the point is uh as you said when when a child goes to a traditional school or whatever I I am personally against grades and notes and all these kind of stuff because they are um there's a bunch of literature that say how bad they are for motivation of the kids but for me they have also another uh side effect first they are delayed in time which means if you think in in real activ inference there are posteriors that never refer to what they should refer to if I have a bad grade of a test that for a test that I've done last week okay I received my error signal doesn't come from the test doesn't come from the mistakes that I've made in the test it comes from the red notes that I received from the teacher what I will try to fix in my environment and how I will try to update my generative model is going to be oh I want to avoid that this is my prediction error my my attempt on the long run to solve uncertainty right but this doesn't fit with the idea that we want the child or the poils or whatever or even the the young adults to learn the content that was in the test in the first place this is the the this is the in my opinion the main big problem the second problem which somehow the mono solved but the inference is better is how we think how we have to design learning because if you think learning as something that is uh that Fs in the brain of the child

it's thinking in active inference terms is in my opinion even more precise than thinking in terms of construction of knowledge right because the knowledge is already somehow constructed you arrive and you try it out the knowledge that you have and then you refine and refine and refine is this approach that would I think could help in design learning environments Learning Materials how we design the books how we design the uh interface how we design the the team group they have to be designed in this way a child that wres think about the the basic I I try to think in very very practical terms uh a child that arrives to school is six and has to learn to read and write and the first thing this is a the mon meod on this it like it's it's it's okay because theoretically you teach literacy at very early age so you make the children play around with letters at the age of three or four they literally play around with big letters but theoretically when you go to school and you want to teach children to uh read and write what do you do the children arrive the first day uh in class and they start with the a tell me how can a cognitive system predict something as such when a child six knows how many words are connected by the the the thing that you want to learn think about how this is complete nonsense in my opinion is complete nonsense instead if you think the other way around what does the child know already the child knows Stories the child know characters the child knows structures the child knows context and you think to use these as form of predictions that the child will make and every time you add one single element of uncertainty manageable uncertainty then it's going to be way more way more natural I'm sorry to say but that's it that's is uh that's is how it is okay I'll read a question next from the live chat glea maximalist wrote what is the monor perspective on peer learning and social Clicks in the classroom how does this relate to active inference modeling of learning okay um the montori approach to learning is it's it's a little bit uh complex so um in the Mont approach Learning and Development they go together and you have the phases of development which in my opinion fits way better with the with the activ inference that the the the the of development of P or bosi they are kind of rough periods of development in which you construct abstract competencies uh but if you think in specific terms monori think that the first competence that the cognitive system uh uh construct is the um capacity of absorbing from the word okay the this is the first thing that a child does monor describes uh the the uh the infant the newborn as searching into the environment the information that he can get it has very much to do with our approach in active inference terms uh of of the model of learning um albite of course is not so precise first for uh um historical reasons uh she didn't have the right terminology she didn't have the the right knowledge uh she didn't have the right uh neuroscientific knowledge and many of the things

that montesori said or claimed they H had have been proved uh uh only in the past 50 years or so there is a beautiful book on mirror Neuroscience written by Leonardo Fiaschi the the guy of the mirror one of the guys of the mirror neurons um and for instance about the role of the hand in cognition and imitation the role of the activities the role of perception they are already all of them inside the mirror model and they very much are in line with our approach our active inference approach to learning and cognition um I'd say that they are compatible okay that would be that I should have started with that they are compatible okay um I hope that answers if it doesn't keep going cool yeah well you made a very provocative comparison with how the llms are trained with a corpus and a little bit of fine-tuning and then basically the content delivery instructional approach where there's a corpus there's the textbooks and the material and then with some fine-tuning we get to what a corpus regurgitating machine and along with that there's also these kind of implicit cultural learnings not just related to the embodiment and the layout in the classroom but things like I should be predicting what everyone else predicts or the next piece of information on my foraging Journey will be provided to me um or there is an expertise hierarchy that's inviolable and all these different kinds of higher order properties that that become learnt through the education system and I think you made a great point which is that active inference is more of the tool that helps us analyze these different cognitive settings and then talk about how they align or not or how they fulfill our preferences or not um it's just uh it's very interesting to use this new lens to bring together many of the the critiques that people have raised about education and I guess how do we move from taking the lens and applying it in a critical fashion to proactively designing new curricula and new experiences for for younger and for older individuals I remember uh at some point I don't I don't remember whom but it was a Critic of Education to say there is just one way to uh to reform education we have just to like to cancel schools that is the best case scenario I'm not so extreme but uh the idea to change the idea to bring something like the T inference within educational settings means first and foremost not thinking about education in terms of curriculum not thinking about education in terms of uh what a student should know at the end of the year or what a teacher should teach in one year but is thinking about a environment in which you Foster options thinking about in active inference terms means uh in active inference a cognitive system what does particularly a new cognitive inst in uh active inference uh um agent uh try looks around simply looks around for gathering information because they might be useful for making uh prediction error accurate prediction errors in the future which how do you translate this into education means offering opportunities means

offering as much as it possible for making better prediction errors right means shifting completely the mindset from what a person a child a poop should know to what we can offer what they can know it's clear that's is for me it's the mindset that I'd like to shift yeah that's very important like thinking about the adjacent possible and the perception and the affordances of each individual and that's also how education and accessibility can come together rather than trying to take like a top- down organizational approach or still again a Content delivery approach just slower or diluted rather than actually like welcoming people's differences um okay Gia maximalist writes what about the social structure of the classroom first question and then okay go for that and then I'll ask the second question no no no h okay social structure and and do you think that grades are beneficial for older children it reinforces that letter grades are the delayed readout of performance like many aspects of adult life maybe we should take off the grades also from adult's life that would be make way easier social structure in in the class in the montori classro the social structure are very much again natural so you have kids that forms groups they can always um work with whosoever they want so they form sort of natural friendships so you work more with the people with whom you work better so to say um and the social structure of course I couldn't go too much into detail um about different social learning but if you think how the uh the uh information is provided Ed in a in a in a Mont classroom the teacher never teaches so you will never find a teacher that tells you some stuff and uh you have to repeat this stuff most of the time particularly for basic competencies the teacher shows how to use certain materials or helps you in uh finding information online things such this kind approaches and montori was incredibly precise and really sometimes when I read her really like I'm astonished it was incredibly precise saying how the teacher should talk with with young children using gestures using certain gestures in a way that they could better vate the the message if I think in terms of social learning and I think about uh five or four years old a six years old and a teacher which is a adult competent individual and I think in terms of social learning what does it do stimulates over imitation which is what is considered the typical the most common of our way to acquire information socially which is also in traditional schooling setting the one that we never use and have you ever tried to copy during a test it's the first the first thing that you're are supposed not to do but it's the most natural way in which we transmit information uh for saying about the social structure this kind of different mechanisms that again according to the inference I see I see lots of advantages to use active inferences for learning because it allows to think the diff kind of modalities in which we learn we transmit and we learn information uh cultur information among each other

uh as a society not as separated in essence but more like a sort of spectrum okay I try alone if I don't succeed then I will look I will shift my attention to something else and so on and so forth in a monor classroom this is allow children then can try out by themselves if it doesn't work they can work collaboratively uh collaboration if I want to talk about not only social structure but also the social learning mechanism that are involved they work collaboratively they imitate each other all the time uh the imitation is not simply um used like in in in in Traditional High School you imitate the cool kid in the corridor for how it's dressed up you do the same in a Mont classroom just you can work with them and usually is not the child that is dressed the better but the child with whom you work better that is your cool child in a classroom this changes a lot the the Dynamics uh grades grades grades grades there is right now there is a big movement against against grades I think they have been published at least four books in the past five six years against grades the use of grades in in uh in at for older kids the problem with grades is that there is a gap between whatever system we have for children in compulsory school and when when we go to university that is the main the main Gap till the university works with exam and grades and and so on and so forth we will have to keep the grades into the high schools because University allows the access based on some numbers that are given to you the later we put the great the better and the reason is very simple very young children they don't um they haven't yet developed the ability to understand to make a difference between what they do and who they are I have a bad note I am a bad person I it's very hard for them to make all these levels these are all kind of refinement of the generative model that comes when you have multiple experiences for which when you are 17 18 19 you discover you start to discover that you might be very bad at doing something but you might be very good at doing something else for younger kids grades means a sort of identity that is the reason why I would push them as back as it possible honestly and there is also an ex extensive literature in education looking at how uh teachers should give grades because they uh are really really really bad for motivations really really bad I can give you a personal experience I was very good in chemistry till at some point I got a bad grade because my teacher discovered that I I wasn't uh doing just my test but I was doing also the test for two of my classmates and instead of saying okay you are very good you have been so good and so fast that you could finish your own test and making the test for other two classmates she gave me a b grade and never been in the class of of chemistry ever again uh and I think as me since I'm not special as me many other kids many other uh students would feel exactly the same yeah another kind of piece with the language models and the synthetic intelligence in the

classroom yes they're disembodied and there is that angle that we explored also though it's kind of like there's a new smart kid on the Block there's a new affordance which is like what year did this happen in and that went from being something maybe that could be found in a book more actions that have to be taken to walk to the library open the book Etc to maybe now search for it online to maybe now it's just going to be like clap your hands and say it again or maybe it's listening the whole time and so how will meta on developing personal agency and a feeling of being in the driver seat in the education be cultivated so people feel like they have tools to access and they have affordances at their disposal rather than feeling I don't know how it could or would feel but to be under the burden of much more knowledgeable much faster agents that are directing the students and teachers I think might also throw up their hands because they're not necessarily going to have recourse for technical questions either and there will be this affordance at hand that will just bring out an essay or increasingly the audio and the visual so how do we have a kind of fusion environments or different zones that help us wisely use that technology that's is a very so I am a Sci-Fi fan Okay so I am not scared about anything of this kind bring in I can deal with that um honestly after having been use having been using for uh at this point one year consistently large language models while the at the beginning I thought okay this is really a game Cher right now I think um not so much at the end um the point is our approach the point is how we teach people how to use them and as for every other piece of technology we all will use a different in different way in different circumstances if I have to uh to translate I live in Germany if I have to translate a piece that I'm writing for sending to I don't know an mail to the mother of uh my daughter of um my daughter friend uh I can use J GPT and I will send the translation as it is okay I won't pay too much attention when I use jpt for working instead not only I check thousands of times the information that I have received because despite it might have faster access to the knowledge but doesn't have the right knowledge somehow and this is something that you feel when it's right and when it's not when you write a piss and you make J P I make J P checking my grammar and my mistakes because I'm not mother pong when gives me back I always pay attention to put just no more than 10 lines because otherwise restructures my sentence and usually what comes out is nonsense is changes completely the meaning and stuff as such so uh yes they are useful we have to learn how to use them as we have learned to use uh to use I remember in 2011 yes was 2011 came out a paper I think it's called the Google effect and they looked at how much we have uh uh in in in the opinion of the of the uh research Searchers we uh we have been losing memory because we started to use Google all the time instead of

memorizing things right uh I see things in a different way uh we haven't lost memory we have learned how to categorize things that we want to remember in a different way I don't have to store the information itself I have to remember how to retrieve that information and the same will be for jpt and for the large language models also sometimes ago I was I was using jpt and at some point and this is the way in which the moment in which it was clear to me um I asked for some uh references about some tool making stuff of for earo and it provided me with a book of uh Kim sery uh um of a book of Kim sney and uh Beyond I know Kim perfectly I know his work very very very nicely and I looked at this book I said but this book doesn't exist I'm 100% sure that this book I I think I I looked at the title I think this is the book that he should write but this book doesn't exist because at the end of the day I know better nice yeah many funny things like that and and knowing where to find the information whether the internal foraging or external Niche modification it's kind of always been the game it's just a little bit modified now um now I'm sorry interrupt you I'm now working on uh um txm on extended mind and learning and the active inference here has a very nice way to uh to change the game of uh extra ST ding of information and so on she's very very interesting yes I have a friend who says he has the largest collection of seashells in the world he leaves them on all the beaches um well where do you go from here or what would you like to see happen or what would you suggest to people who are excited about pursuing education and active inference uh I don't know because I I as I said I I tried to to to create with jpt I would I would show to you all the attempt uh to create a um image um of a classroom in which was applied the theory of cognition of activ infant with the Mont method and I had lots of weird results because it's something that apparently nobody has tried yet but what I would suggest is trying to make uh practical to think in terms of design to think in terms of affordances to think in terms of everyday interactions to think in terms of buildings to think in terms of artifacts this I think is going to be the The Next Step awesome yeah just to read a few comments Joanne wrote it brings the environment into the picture we need to examine that more closely collaboration not competition and then Jr wrote It's like lifting memory from remembering the object to remembering the process in a more complex context as the world evolves yes yes the the the good aspect I think the point of putting together something like the monor meod and active inference is understanding that learning is not about acquiring information but about understanding the information how to use the information that is the the the game changer or as my daughter says all the time Mama today I invented mathematics amazing well Laura thank you very much for the work it was definitely a topic that people I feel like wanted and

want and need to explore how to be active in education and not get swamped in the tools and in all of these efforts and to kind of keep the hope alive for what education can be yes yes if you should put a little bit more uh ecology and sustainability inside and then we have the complete picture awesome well till next time thank you very much next time thank you very much bye bye